

Post-Quantum Cryptography Migration

Part 11 - Where It Runs: Cloud Infrastructure

Compiled by the Spaced Research Ledger

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Most of the world's computing now runs in a handful of clouds, so a single provider decision moves more traffic to post-quantum than a thousand enterprise projects. The providers know it, and they have turned the migration into a competitive race with published roadmaps and dated milestones.

This report covers the field in seven sections. Sections 1 through 3 cover AWS, Azure, and Google Cloud. Section 4 covers the other providers, Section 5 the content-delivery networks, Section 6 the secure-access services, and Section 7 the attestation chains inside confidential computing.

The standings are uneven. AWS turned hybrid key exchange on by default across its core services and put ML-DSA into its key and certificate services. Google matched that and became the only major cloud with all three NIST algorithms generally available in its KMS. Azure trails visibly, with its own documentation claiming no post-quantum key types at all.

The edge moved fastest of anyone. Cloudflare and Akamai now run post-quantum end to end, and both have committed to full protection, signatures included, by 2029. At the other extreme sits confidential computing, where every attestation chain in every cloud still rests on classical elliptic curves.

1. AWS

Amazon's cloud secures more workloads than any other, and its migration is organized as a program: inventory first, public endpoints second, roots of trust third. What AWS operates, it upgrades itself. What customers configure, they must switch on. This section covers how far each half has come.

Recent Developments

The FIPS gap closed. AWS KMS now runs hybrid post-quantum TLS on all endpoints, the FIPS-validated ones included [1]. Two absences are worth naming precisely. No primary source describes shipped post-quantum features in the Nitro virtualization system, where commentary is forward-looking only [2]. And nothing confirms the managed Site-to-Site VPN supports a hybrid key exchange natively [3].

Current State

The default flipped in spring 2026. AWS turned ML-KEM hybrid TLS on by default across KMS, Secrets Manager, ACM, Payment Cryptography, and S3 [4]. The lineage runs back to 2022, when the KMS endpoints offered three draft-era KEMs [5], narrowed to Kyber in 2023 [6], then standardized. The cost is measured: about 1,600 extra handshake bytes and well under a millisecond [7]. The foundation is AWS-LC, the first open-source module with ML-KEM in a FIPS validation [7].

The customer-configured surfaces are opt-in. Load balancers support post-quantum TLS policies everywhere at no cost, but listeners must select them [8][8], with three hybrid groups and named recommended policies [9][9]. CloudFront went the other way, enabling hybrid key establishment automatically on viewer connections [10] and adding a TLS 1.3-only policy [10]. It earned exclusion from AWS's own readiness scanner because there is nothing left to configure [11]. The scanner covers the surfaces where policy choice still matters [11].

Signatures run through the certificate services. Private CA issues ML-DSA for roots of trust spanning code signing, device authentication, and tunnel authentication [12][12]. Public ACM certificates remain classical [13], and the original transport rollout never touched certificate content [5]. The program frames it all as workstreams under a shared-responsibility split: AWS endpoints prefer post-quantum automatically, customer resources need configuration [12][12].

The VPN gap illustrates the split's edge: the available mechanism is post-quantum-signed certificates for tunnel authentication, not a post-quantum exchange inside the VPN protocol [12][3][12]. Nitro's baseline, meanwhile, is a clean public record with no known compromise, ahead of any quantum-specific upgrade [14].

2. Azure

Microsoft's cloud inherits its cryptography from Windows, which means Azure's post-quantum story is largely the Schannel and SymCrypt story told elsewhere. What Azure has not done is put the new algorithms into its key-custody services, and its own documentation is the bluntest witness. This section covers the gap between platform and product.

Recent Developments

The key services remain classical by the best available reading. A third-party analysis, itself marked as needing verification, found Key Vault supporting only RSA, elliptic-curve, and AES keys, with no FIPS 203, 204, or 205 algorithm available [15]. Managed HSM matches, with ML-KEM and ML-DSA explicitly unsupported and the underlying hardware validated to the older FIPS standard [15][15]. The workaround is imported keys from external hardware, which puts post-quantum material under Azure's access control without giving it HSM-attested operations [15].

Current State

Microsoft's own pages say the quiet part. The Key Vault documentation claims quantum resistance only for 256-bit symmetric keys, framed around Grover's algorithm, never a NIST asymmetric algorithm [16]. Managed HSM's documentation echoes it exactly [17]. The shipped movement is symmetric too: AES key wrapping reached Key Vault Premium in preview, extending quantum-resistant wrapping [18].

Against that stand two uncorroborated third-party claims [19]. One says ML-KEM and ML-DSA arrive through SymCrypt by tier and region, the other that TLS to the Key Vault endpoint is progressively hybrid [19]. Neither has a Microsoft source behind it.

The edge is similarly unannounced. A tracking site's Front Door assessment contradicts itself, saying in one place that no customer toggle exists and in another that a rollout has begun [20]. No Microsoft statement supports either version [20]. Application Gateway follows the Windows Schannel cadence rather than any gateway-specific release [20].

The practical guidance is telling: terminate TLS on a customer-controlled backend and run the gateway as a pass-through [20]. Architects are told to build health checks that log negotiated suites, because the default cannot be assumed [21].

3. Google Cloud

Google ran post-quantum cryptography in production before NIST finished standardizing it, on the internal protocol that encrypts traffic between its own data centers. Its cloud now publishes the most specific roadmap of the three: migration complete by 2029, a year ahead of the government's deprecation clock. This section covers the provider setting the pace.

Recent Developments

The roadmap keeps its dates. Quantum-safe key import reached preview in Cloud KMS in August, wrapping imported keys with ML-KEM or X-Wing in transit [22], on schedule against the published plan [23]. The internal protocol's next step is authenticity, adding post-quantum signatures to the ALTS handshake to gain operational experience with post-quantum certificate chains [24], targeted for 2026 to 2027 [23]. The load balancers got the most explicit rollout in the industry: opt-in now, default with an opt-out from October 2026, and permanently on with no opt-out after October 2027 [25].

Current State

Cloud KMS leads the market on algorithms. It is the only major cloud KMS with all three NIST families generally available as native key types, with pre-hash variants chosen deliberately to make large-payload signing practical [26][26]. A dashboard chart now shows customers which asymmetric keys remain quantum-vulnerable [27].

The history explains the confidence. Google enabled a conservative pre-NIST KEM on ALTS in November 2022 [28] and migrated it to standardized ML-KEM as a completed milestone [29]. The company target is full migration by 2029, with store-now-decrypt-later mitigated by 2027 [29]. Its public endpoints already offer hybrid key exchange to all comers [30]. A hands-on account quantifies the wire cost, with hello messages growing several-fold, a fragmentation concern but not a practical problem for browsing [31].

The load-balancer rollout requires TLS 1.3 by construction [32]. The honest caveat is hardware. The 2028 target for HSM and confidential-computing roots of trust is Google's own, and the roadmap admits equipment cycles could push parts past 2029 [29].

4. Other Cloud Providers

Beyond the big three, the cloud market includes providers with their own migrations: Oracle, IBM, Alibaba, and regional players. Their timelines range from earlier than anyone to documentation that may be years stale. This section covers the rest of the field, where the sharpest single product is a database.

Recent Developments

Oracle put the algorithms in the database itself. Oracle Database 19c supports ML-KEM, ML-DSA, and SLH-DSA under its next-generation provider, in non-FIPS and FIPS 140-3 modes but not the legacy FIPS mode [33].

Current State

Oracle's push is broad and dated. Its security team implemented ML-KEM and ML-DSA in its providers as the first step toward hybrid key establishment across TLS, SSH, and IKEv2 [34]. Its flagship database added pure ML-KEM key exchange in October 2025 and hybrid in January 2026, claiming a first among database systems [35]. The FIPS-mode boundaries are documented [36], and newer releases default stronger tablespace ciphers [37][37].

Partners fill the network layer, with a post-quantum VPN in its government regions [38] and cryptographic-discovery tooling for defense customers [39]. Java workloads ride the JDK roadmap toward its September completion [40].

IBM moved earliest and may have aged worst. It announced quantum-safe cloud services in November 2020, well before standards [41]. Current documentation still describes round-three Kyber parameters and round-two Dilithium, and it is unclear whether that reflects deployed algorithms or stale pages [42][42]. The broader Quantum Safe branding spans mainframes and consulting, context rather than cloud capability [43].

Alibaba matched the default-on edge pattern. Its edge-acceleration service enables the hybrid group sitewide with no configuration [44]. It offers a beta post-quantum HSM covering the NIST set plus the hash-based schemes [45]. It claims a first with post-quantum algorithms in vehicles through an XPeng partnership, self-reported [46]. Ambitions run from chips to flying cars [46], with a research lab in the background [47].

5. CDN & Edge

The content-delivery networks sit between every browser and most large websites, which makes them the internet's fastest lever: one CDN default protects millions of sites at once. Two networks account for most post-quantum TLS on the internet today. This section covers the layer that finished first.

Recent Developments

Akamai completed the loop. In August it reached general availability for end-to-end post-quantum protection across its network, client-to-edge on by default and origin plus internal legs complete [48]. It is evaluating its roadmap against the industry's 2029 target, certificates included [49]. Cloudflare shipped the harder half early: post-quantum origin authentication with ML-DSA went live in June, weeks before its announcement

[50][51]. The rollout had one incident, a certificate-validation change that briefly invalidated a small number of customer certificates [50].

Current State

Cloudflare is the reference deployment. It has worked on post-quantum since 2017, enabled client-side hybrid key agreement for all customers in 2022, and reached origin key agreement in 2023 [52]. Its current status page is precise [53][54]. Key agreement is on in both directions, and origin signatures are on through its authentication features.

Visitor-side signatures wait on Merkle Tree Certificates, and internal migration is mostly done with a long tail [53][54]. The engineering details are candid, including a library fork four years behind upstream and a patched interim implementation while Go lacked ML-DSA [55], with documentation current [56].

Its accelerated 2029 commitment cites the recent resource-estimate research and covers every plan tier free [57][57], with dated milestones [58]. Its essays argue ML-DSA will have to do rather than waiting for better algorithms [59], noting a U.S. executive order's 2031 deadline [59].

Akamai ran the same race with opposite instincts. It positioned itself as a standards-follower, waiting for the IETF and libraries [60]. It then executed a three-phase rollout from origin to edge to internal [61], building its own provider rather than waiting on upstream [62]. Its origin leg requires TLS 1.3 and degrades gracefully by one round trip [63].

Fastly announced its rollout in April 2025 with little since [64]. Yet the measured result stands: Fastly and Cloudflare together account for nearly 70 percent of observed post-quantum TLS deployments [65].

6. SASE & Zero Trust

Secure-access platforms sit inline on enterprise traffic, decrypting, inspecting, and re-encrypting it. That position makes them natural translators: a proxy can speak post-quantum to the world on behalf of applications that cannot. This section covers the vendors racing to be that translator, and how far each has actually shipped.

Recent Developments

The proxies are lighting up in sequence. Netskope's secure web gateway added beta hybrid key exchange on both the client and server sides of its proxy, support-enabled rather than self-service, key exchange only [66] [66]. AT&T and Palo Alto Networks announced a quantum-resilient SASE fabric integrating the NIST algorithms into SD-WAN appliances over a private backbone [67]. It is built on the IETF's IPsec mechanisms, with trials in summer 2026 [68]. Zscaler's platform extends protection across the last mile even when destination servers lack support [69], and paired with a consulting partner for discovery services [69].

Current State

Zscaler claims the inline-inspection first. It decrypts hybrid post-quantum traffic, applies full content inspection, and re-encrypts onward, styling itself a quantum-safe intermediary [70]. It supports the

preshared-key IPsec mechanism for branch connections [71] and anchors its positioning to the June 2026 U.S. executive orders and their 2030 and 2031 deadlines [70].

Palo Alto shipped the deepest protocol support. Its SD-WAN release implements the preshared-key and multiple-key-exchange RFCs with up to seven rounds, plus the standard hybrid TLS groups for control connections [72]. Its firewalls offer cipher translation, quantum-safe outside and classical inside, framed as a compatibility bridge for unupgradable applications [73]. Uniquely among vendors here, it exposes non-selected experimental KEM families as a labeled option [73]. Its remote-access agent mirrors the same mechanisms [74].

Netskope is earlier stage and says so. Its public work centers on an architecture white paper mapping five encryption points [75]. The quantum-resilient platform remains pre-availability months after announcement [76], with executive framing about 2026 being the year of how [77]. Market analysts rank it a SASE leader anyway, with post-quantum readiness one factor among several [78].

7. Confidential Computing Attestation

Confidential computing promises that even the cloud provider cannot see inside a running workload, proven by hardware-signed attestation reports. Every one of those signatures is classical elliptic curve. This section is short because the work has not started, and that absence is the finding.

Current State

Every chain is classical, uniformly. AMD's attestation reports are signed by per-chip elliptic-curve keys [79]. They root in an all-ECDSA certificate chain confirmed across AMD, AWS, Azure, and VMware documentation, with no post-quantum algorithm mentioned anywhere [80]. Intel's TDX quotes run through an ECDSA attestation key certified back to hardware fuses [81], and its verification library exposes no post-quantum option [82].

AWS's enclave attestation shows the structural problem concretely. Its documents are signed with ECDSA under a 30-year root [83], and the key-delivery mechanism supports only RSA encryption [84]. An open issue requesting post-quantum support notes that an ML-KEM public key physically does not fit the document's fixed field, requiring a format change [85]. The hardware roots of trust, everywhere, are the migration's last mile.

About This Report

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